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# Low molecular weight and highly functional RCF lignin products as a full bisphenol A replacer in bio-based epoxy resins†

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**Herein, we present a full lignocellulose-to-chemicals valorization chain, wherein low molecular weight and highly functional lignin oligomers, obtained from reductive catalytic fractionation (RCF) of pine wood, were used to fully replace bisphenol A (BPA) for synthesizing bio-based epoxy resins.**

Epoxy resins represent an important class of thermosetting resins, generally possessing excellent adhesion properties, chemical resistance and good performance at elevated temperatures. They are employed in diverse applications such as coatings, adhesives and composite materials. However, epoxy resins are typically based on toxic and petroleum-based aromatic precursors, such as bisphenol A diglycidyl ether (BADGE) and are hard to recycle.<sup>1–3</sup> Therefore, it is important to use a source of renewable aromatics to synthesize epoxy resins without BADGE.<sup>2,3</sup> In this respect, lignin is an ideal renewable aromatic carbon source candidate.<sup>2,3</sup> Widely available technical lignins (*e.g.* Organosolv, Kraft) or fractions thereof have been studied as partial or full replacement of petrochemicals for material applications, often reporting interesting properties dependent on lignin structure, functionality and molecular weight ( $M_n$ ). For example, Nejad *et al.* recently found that BPA could be fully replaced in the synthesis of epoxy resins by choosing the right lignin fraction.<sup>4</sup> Their results showed that technical lignins with higher phenolic content and lower  $M_n$  and dispersity index ( $\mathcal{D}$ ) were most suitable for replacing 100% of BPA in the formulation of epoxy resins.<sup>4</sup> Given this conclusion, the use of RCF lignin oligomers (Supplementary note 1, ESI†) which is known to have a low  $M_n$ , low  $\mathcal{D}$ , and high phenolic

content, though never tested,<sup>5,6</sup> should be very promising for the synthesis of BPA-free epoxy resins.

A plot of the phenolic and aliphatic hydroxyl content, and  $\mathcal{D}$  of a variety of (technical) lignins and RCF lignin relative to their  $M_n$  reveals the unique properties of RCF lignin (Fig. S2A–C, ESI†). Fractionation of technical lignins is commonly used to improve lignin properties.<sup>7,8</sup> Even though fractions with lower  $M_n$  and high phenolic content can be obtained, RCF lignin fractions exhibit a similar or higher phenolic content in combination with a lower  $M_n$ . Besides, such fractionation strategy starting from technical lignins valorizes only part of the lignin stream leaving behind significant amounts of a high molecular weight lignin fraction with low functionality.<sup>7,8</sup> This is in stark contrast with the low molecular weight RCF lignin which is composed of highly functional monomers and low  $M_n$  oligomers.<sup>5,9</sup> These monomers can be selectively extracted and utilized towards various commodity and specialty chemicals (phenol and propylene production,<sup>10</sup> less toxic bisphenols,<sup>11</sup> terephthalic acid,<sup>12</sup> catechol<sup>13</sup> and pharmaceutical compounds<sup>14,15</sup>), leaving behind the lignin fractions with low  $M_n$  oligomers, high functionality and low  $\mathcal{D}$  (Fig. S2A–C, ESI†). A few groups recently used the RCF lignin oligomers in proof-of-concept application development. Tan *et al.* reported the application of RCF lignin oligomers as partial replacement of polyols in the synthesis of polyurethanes.<sup>16</sup> Torr *et al.* showed that a hydrogenolysis lignin oil or a part of the hydrogenolysis oligomers can be used as a partial replacement for BPA in the synthesis of epoxy resins.<sup>17–19</sup> However, a full BPA replacement was unsuccessful in their study. Recently, successful replacement of *p*-nonylphenol by RCF lignin oligomers in the synthesis of resin and varnish used in printing inks has been demonstrated.<sup>10</sup> Importantly, a recent study showed the first steps towards a large scale RCF process,<sup>20</sup> and a life cycle assessment showed a net positive carbon footprint for the production of these RCF lignin oligomers.<sup>10</sup> Given the promising properties of RCF lignin oligomers for the production of epoxy resins,<sup>4</sup> we herein study their glycidylation and use as a full BPA replacer in the synthesis of epoxy resins. A general

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† Electronic supplementary information (ESI) available: Detailed experimental procedures, supplementary notes on RCF, separations, <sup>1</sup>H–<sup>13</sup>C HSQC spectroscopy, reaction optimization and thermal analysis; additional GPC chromatograms, <sup>1</sup>H-NMR spectra, <sup>31</sup>P-NMR spectra, <sup>1</sup>H–<sup>13</sup>C HSQC spectra, FT-IR spectra and additional figures and tables. See DOI: 10.1039/d1cc02263f

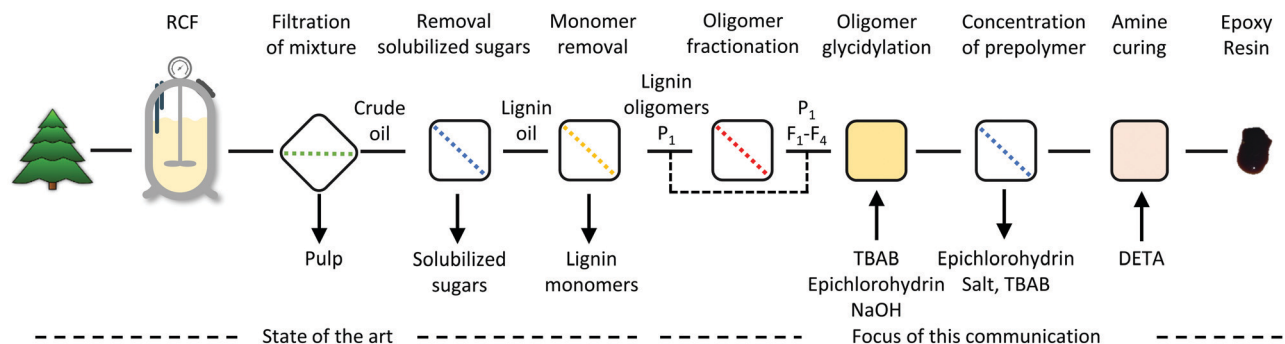


Fig. 1 General overview of process steps to obtain the final lignin based epoxy thermoset.

overview of this communication is presented in Fig. 1. In a first step, RCF lignin oil is produced using a Ru/C catalyst, enabling easy lignin monomer removal according to state-of-the-art procedures (see Supplementary note 1, ESI†), resulting in a RCF oligomer fraction (Parent fraction;  $P_1$ ).

Subsequently, thorough RCF oligomer characterization is combined with sequential fractionation based on a novel solvent anti-solvent precipitation, in contrast to earlier solvent extraction methods,<sup>5</sup> to study the impact of  $M_n$  and lignin structure on the next steps of the valorization chain (see Supplementary note 2, ESI†).<sup>5</sup> In total, four distinctly different fractions of RCF oligomers ( $F_1$  to  $F_4$ ) are obtained. Next, all five fractions ( $P_1$ ,  $F_1$ – $F_4$ ) are glycidylated to obtain epoxy prepolymers, which are cured with an amine linker to obtain epoxy resins and their thermal properties are investigated.

As reported in Table 1 and Fig. S6 (ESI†), the entire RCF oligomer fraction  $P_1$ , has a  $M_n$  of 703 g mol<sup>−1</sup> and  $M_w$  of 985 g mol<sup>−1</sup>. Separation of  $P_1$  resulted in fractions  $F_1$  to  $F_4$  with a  $M_n$  decreasing in this respective order from 1146 to 446 g mol<sup>−1</sup> and a  $M_w$  decreasing in the same order from 1781 g mol<sup>−1</sup> to 541 g mol<sup>−1</sup>. As a result,  $\bar{D}$  is very low which is a good property to serve as an epoxy substrate. Besides, these fractions all have  $M_n$  more than double that of a lignin monomer (*i.e.* between 150 and 220 g mol<sup>−1</sup>), indicating a potentially interesting substrate for glycidylation and incorporation in an epoxy thermoset since two or more phenolics per substrate molecule are necessary for final crosslinking.<sup>1,3</sup> On the other hand,  $M_n$  is not too large, which otherwise could be an issue as

demonstrated in the synthesis of epoxy resins from high  $M_n$  technical lignins.<sup>4</sup> As reported in Fig. S7 (ESI†) for  $P_1$  and Fig. S8 (ESI†) for  $F_1$ – $F_4$ , many lignin inter-unit linkages and end-units resulting from the reactions during RCF can be detected by <sup>1</sup>H–<sup>13</sup>C HSQC spectroscopy. Hydrogenolysis of most of the native lignin ether linkages and their selective conversion to RCF analogues is proven by <sup>1</sup>H–<sup>13</sup>C HSQC spectroscopy (Supplementary note 3, ESI†). In addition, aliphatic end-units – the hydrogenolysis products – are found to a much higher extent (Supplementary note 3, ESI†). This is in accordance with a recent study on Pd/C-catalyzed RCF.<sup>5</sup> Note the selectivity difference between the 4-propyl and 4-propanol groups in the Ru/C RCF lignin fractions used in this work and the Pd/C RCF lignin fractions in our previous work. The high number of aliphatic end-units is important because it improves processability and solubility of the lignin, owing to the absence of C–C condensation reaction and concurrent  $M_n$  increase. Furthermore, the chemistry during RCF also led to a high content of phenolic groups, which are most reactive for formation of oxirane moieties. The amount and nature of these reactive phenolic groups is determined by <sup>31</sup>P-NMR, a frequently applied method in lignin analytics.<sup>5</sup> As reported in Table 1, the total phenolic content of  $P_1$  is 4.2 mmol g<sup>−1</sup>. Approximately 42% of these phenols are substituted on their aromatic C<sub>5</sub> position (Table S3, ESI†), because of their native β-5, 4-O-5 and 5-5 interphenolic linkages. A general trend in  $F_1$ – $F_4$  is the increase of the phenolic content with decreasing  $M_n$ , attributed to the presence of small amounts of unreacted β-O-4 linkages and the increase of uncleaved 4-O-5 linkages in higher  $M_n$  fractions (*viz.*  $F_1$  and  $F_2$ ). Overall, the phenolic OH content is high in all fractions, resulting in an average of more than 2 phenolic OH groups per RCF lignin molecule in each fraction, rendering all of them potentially valuable for glycidylation. Note that the RCF lignin used in this communication possesses a unique combination of low  $M_n$ , low  $\bar{D}$  and high (phenolic and aliphatic) OH-content compared to the variety of other (technical) lignins (fractions) used in earlier epoxy resin synthesis as displayed in Fig. S3A–D (ESI†). In the next step, oxirane moieties were introduced in all five fractions by glycidylation of the phenolic group. To avoid any undesired side reactions (*viz.* ring opening or polymerization), a two-step protocol was optimized (Supplementary note 4, ESI†). In accordance with literature, a high selectivity to oxirane moieties was obtained for the RCF lignins by carrying out the first step at higher temperature (85 °C) and the second step at

Table 1 Characteristics of the RCF oligomers  $P_1$  and its derived fractions  $F_1$  to  $F_4$ , measured by GPC and <sup>31</sup>P-NMR

|       | Yield <sup>a</sup> (%) | $M_n^b$ (g mol <sup>−1</sup> ) | $M_w^b$ (g mol <sup>−1</sup> ) | $\bar{D}^c$ | Total phenols <sup>d</sup> (mmol g <sup>−1</sup> ) |
|-------|------------------------|--------------------------------|--------------------------------|-------------|----------------------------------------------------|
| $P_1$ | 46                     | 703                            | 985                            | 1.40        | 4.20                                               |
| $F_1$ | 3                      | 1146                           | 1781                           | 1.55        | 3.56                                               |
| $F_2$ | 17                     | 898                            | 1212                           | 1.35        | 4.04                                               |
| $F_3$ | 12                     | 662                            | 814                            | 1.23        | 4.20                                               |
| $F_4$ | 14                     | 445                            | 541                            | 1.22        | 4.37                                               |

<sup>a</sup> The yield is based on the total amount of DCM-soluble oil, extracted from pine during the RCF process. <sup>b</sup> Quantified by GPC measurements. <sup>c</sup>  $\bar{D}$  = dispersity index =  $M_w/M_n$ . <sup>d</sup> Quantified by <sup>31</sup>P-NMR (Fig. S9–S13, ESI).

room temperature.<sup>4</sup> Interestingly, given the excellent solubility of RCF lignin, no addition of a co-solvent was necessary which is often the case for lignin glycidylation.<sup>4,7</sup> <sup>1</sup>H-<sup>13</sup>C HSQC spectroscopic analysis confirmed introduction of the oxirane moiety on the aromatic part of the lignin structure (Fig. S16 and S17, ESI†).

The number of oxirane moieties is expressed as the epoxy content (in mmol g<sup>-1</sup>) or the epoxy equivalent weight (EEW; g eq<sup>-1</sup>). These values were determined by <sup>1</sup>H-NMR, taking into account the overlapping H-groups from the lignin backbone (Fig. S18–S22, ESI†).<sup>4,7</sup> Overall, an epoxy content between 3.44 and 3.75 mmol g<sup>-1</sup> was detected, resulting in an EEW between 267 and 291 g eq<sup>-1</sup> (Table 2). Next to the experimental values, the theoretical epoxy content can be calculated as well based on the number of phenols in the starting fractions, the percentage of unreacted phenols (based on <sup>31</sup>P-NMR) and the mass increase by the introduction of the oxirane group. Theoretical epoxy contents between 2.96 and 3.50 mmol g<sup>-1</sup> were calculated (Table 2, Theoretical epoxy content 1). Thus, higher experimental epoxy contents were observed than theoretically calculated. This is because a portion of the aliphatic OH groups also reacted with epichlorohydrin to form oxirane groups, as supported by the reaction (under similar conditions) of the model compound, 3-phenyl-1-propanol, and <sup>1</sup>H-<sup>13</sup>C HSQC spectroscopic analysis on the lignin fractions (ESI† Supplementary note 4 and Tables S6, S7). This shows that up to 20% of the oxirane groups are bound on aliphatic lignin functionalities, which is a significant contribution of the aliphatic hydroxyl groups, and in contrary to previous findings.<sup>4</sup> Taking into account the reactivity of these aliphatic hydroxyls, the experimental epoxy content closely matches the theoretical one (Table 2, theoretical epoxy content 2). The results also reveal that the efficiency of incorporation of the oxirane group is irrespective of *M<sub>n</sub>*, illustrated by the low number of residual phenolic OH groups for all fractions. Another important property of the epoxy pre-polymers is their *M<sub>n</sub>*. Both the introduction of oxirane groups and side reactions (*i.e.* etherification) can increase molecular weight. As expected *M<sub>n</sub>* and *M<sub>w</sub>* of the glycidylated fractions increases (Fig. S23, S24, ESI† and Table 2), but this increase is not excessive, is similar for all fractions (Fig. S25, ESI†), and can mainly be ascribed to the incorporation of oxirane moieties. To show the added value of the obtained RCF lignin-derived epoxy pre-polymers, plots of EEW to *M<sub>n</sub>*, *D*, and phenolic content of the RCF-lignin

pre-polymers and a variety of technical lignin pre-polymers was constructed (Fig. S4, ESI†). Clearly, the epoxy pre-polymers synthesized in this work possess a significantly lower *M<sub>n</sub>*, *D*, and a lower EEW compared to other lignin-derived pre-polymers. The RCF lignin pre-polymers thus exhibit unique properties to serve as a valuable feedstock for epoxy resins.

To assess the potential of these different lignin-derived epoxy pre-polymers, all fractions (*P*<sub>1</sub> and *F*<sub>1</sub>–*F*<sub>4</sub>) and commercial bisphenol A diglycidylether (BADGE) were cross-linked with commercial diethylenetriamine (DETA). To avoid excessive cross-linking, which might result in brittle resins,<sup>1,18</sup> 0.33 mole of DETA was used per mole of epoxy. Effective curing is confirmed by the disappearance of the 912 cm<sup>-1</sup> vibration band – related to the epoxy ring<sup>21</sup> – in the cured samples, indicative of successful conversion of epoxy groups (Fig. S26 and S27, ESI†). The thermal properties of the obtained epoxy-amine resins were determined after curing, as shown in Table 3. Thermogravimetric analysis (TGA) was performed to evaluate the thermal stability of the cured BADGE and the *P*<sub>1</sub> and *F*<sub>1</sub>–*F*<sub>4</sub> thermosets. Fig. S28 (ESI†) shows a one-step degradation TGA curve for all epoxy thermosets under a nitrogen atmosphere, due to the degradation of the cross-linked polymer network.<sup>22</sup> The main characteristics of the thermal degradation are summarized in Table 3. *P*<sub>1</sub> and *F*<sub>1</sub>–*F*<sub>3</sub> show a similar onset degradation temperature (*T*<sub>d,5%</sub>) at around 270 °C, which is approximately 30 °C lower than the *T*<sub>d,5%</sub> of the BADGE resin. Thus, the thermal stability of the *P*<sub>1</sub> and *F*<sub>1</sub>–*F*<sub>3</sub> resins can be presumed to be similar. The lower *M<sub>n</sub>* fraction *F*<sub>4</sub> has a considerably lower thermal stability, *viz.* *T*<sub>d,5%</sub> of 240 °C. This might be due to the presence of a small number of monomers, resulting in a less cross-linked resin. A too low *M<sub>n</sub>* of the lignin fractions due to the presence of monomers therefore negatively impacts the resin quality. Interestingly, the temperature at 50% weight loss (*T*<sub>d,50%</sub>) is distinctly different between all five thermosets and more comparable to the *T*<sub>d,50%</sub> of the BADGE resin. Higher *M<sub>n</sub>* lignin pre-polymers result in a higher *T*<sub>d,50%</sub> of the final epoxy resin whereas epoxy resins based on lower *M<sub>n</sub>* pre-polymers loses mass more rapidly. Possibly, this is because a lower extent of cross-linking – due to the lower number of oxirane moieties per molecule – results in less bonds to break per unit (Fig. S30, Supplementary note 5, ESI†).<sup>4</sup> A similar trend is observed between *M<sub>n</sub>* of the lignin fraction and their

**Table 2** Theoretical and experimental properties of the epoxy pre-polymers based on *P*<sub>1</sub> and its derived fractions *F*<sub>1</sub> to *F*<sub>4</sub>. The experimental properties were determined by GPC, <sup>1</sup>H-NMR, and <sup>31</sup>P-NMR

|                       | Total phenols <sup>a</sup><br>(mmol g <sup>-1</sup> ) | Unreacted phenols <sup>b</sup> (%) | Theoretical epoxy content 1 <sup>c</sup> (mmol g <sup>-1</sup> ) | Theoretical epoxy content 2 <sup>d</sup> (mmol g <sup>-1</sup> ) | Epoxy content <sup>e</sup> (mmol g <sup>-1</sup> ) | EEW (g eq <sup>-1</sup> ) | <i>M<sub>n</sub></i> <sup>f</sup> (g mol <sup>-1</sup> ) | <i>M<sub>w</sub></i> <sup>f</sup> (g mol <sup>-1</sup> ) | Oxirane/molecule <sup>g</sup> |
|-----------------------|-------------------------------------------------------|------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------|---------------------------|----------------------------------------------------------|----------------------------------------------------------|-------------------------------|
| <i>P</i> <sub>1</sub> | 4.21                                                  | 0.4                                | 3.40                                                             | 3.70                                                             | 3.71                                               | 270                       | 895                                                      | 1500                                                     | 3.3                           |
| <i>F</i> <sub>1</sub> | 3.60                                                  | 1.5                                | 2.96                                                             | 3.37                                                             | 3.49                                               | 287                       | 1604                                                     | 2957                                                     | 5.6                           |
| <i>F</i> <sub>2</sub> | 4.09                                                  | 1.5                                | 3.28                                                             | 3.60                                                             | 3.44                                               | 291                       | 1175                                                     | 1934                                                     | 4.0                           |
| <i>F</i> <sub>3</sub> | 4.29                                                  | 2.8                                | 3.38                                                             | 3.54                                                             | 3.75                                               | 267                       | 812                                                      | 1193                                                     | 3.0                           |
| <i>F</i> <sub>4</sub> | 4.41                                                  | 1.2                                | 3.50                                                             | 3.52                                                             | 3.52                                               | 284                       | 511                                                      | 715                                                      | 1.8                           |

<sup>a</sup> Quantified by <sup>31</sup>P-NMR of the lignin fractions (Fig. S9–S13, ESI). <sup>b</sup> Quantified by <sup>31</sup>P-NMR after glycidylation (Fig. S9–S13, ESI). <sup>c</sup> Calculated by taking into account the mass increase after glycidylation of the phenolics. <sup>d</sup> Calculated by taking into account the mass increase after glycidylation of the phenolics and the aliphatic OH as determined in Table S6 (ESI). <sup>e</sup> Quantified by <sup>1</sup>H-NMR. <sup>f</sup> Quantified by GPC measurements. <sup>g</sup> Oxirane/molecule = *M<sub>n</sub>*/EEW.

**Table 3** Comparison of properties of cured epoxy thermosets including TGA data of temperatures at 5% ( $T_{d,5\%}$ ), 30% ( $T_{d,30\%}$ ), and 50% ( $T_{d,50\%}$ ) weight loss, statistic-heat resistant index temperature ( $T_s$ ), char residue and glass transition temperature ( $T_g$ )

|       | $T_{d,5\%}^a$<br>(°C) | $T_{d,30\%}^a$<br>(°C) | $T_{d,50\%}^a$<br>(°C) | $T_s$ (°C) | Char <sup>ab</sup><br>(wt%) | $T_g^c$ (°C) |
|-------|-----------------------|------------------------|------------------------|------------|-----------------------------|--------------|
| $P_1$ | 272                   | 317                    | 342                    | 146        | 22.5                        | 88           |
| $F_1$ | 267                   | 324                    | 384                    | 148        | 31.8                        | 123          |
| $F_2$ | 277                   | 320                    | 358                    | 148        | 28.7                        | 110          |
| $F_3$ | 271                   | 320                    | 346                    | 147        | 23.4                        | 104          |
| $F_4$ | 240                   | 306                    | 328                    | 137        | 13.6                        | 52           |
| BADGE | 303                   | 349                    | 360                    | 162        | 8                           | 61           |

<sup>a</sup> Under  $N_2$  flow. <sup>b</sup> Residue at 575 °C under  $N_2$  flow. <sup>c</sup> Determined by DSC.

char formation and glass transition temperature ( $T_g$ ) (Fig. S30, ESI†).

A higher  $T_g$  reflects the higher extent of cross-links, since increasing covalent cross-links prevents the mobility of polymer segments.<sup>23</sup> Similarly, a higher char content is explained by this higher degree of cross-links,<sup>22</sup> as can be observed in Fig. S30 (ESI†) for the RCF lignin-based epoxy resins. Interestingly, this allows for tuning the thermal properties of the epoxy thermoset by varying the RCF lignin  $M_n$ . Although we can conclude that the thermal stability differs from that of conventional BADGE epoxy resin, RCF-lignin epoxy resins' thermal decomposition occurs at significantly higher temperatures than  $T_g$  and thus should not affect the resin at relevant usage temperature.

Herein, we have delivered a proof of concept of the use of RCF lignin oligomers – having a favourably high phenolic content and low  $M_n$  and  $D$  – as precursor in the synthesis of bio-based epoxy resins. We have shown that RCF oligomers can be fractionated into fractions with distinct molecular properties by a novel sequential solvent anti-solvent precipitation approach. Efficient incorporation of epoxy moieties was demonstrated for all fractions, irrespective of their  $M_n$ . Curing of the glycidylated lignin fractions with DETA resulted in thermosets, with the higher  $M_n$  lignin epoxy resins showing higher  $T_g$ ,  $T_{d,50\%}$ , and char content. The use of the entire oligomer fraction also exhibited adequate thermal properties. Future research should focus on (i) expanding the product scope by using other types of RCF lignin or curing agents in combination with comprehensive mechanical properties characterization of the final epoxy resins, (ii) putting it to the test on larger scale in a commercial application in combination with an extensive TEA and LCA, and (iii) screening the toxicological profile of the RCF lignin pre-polymers.

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## Conflicts of interest

There are no conflicts to declare.

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